

Local Model Bake-Off — ELI on an 8 GB consumer GPU

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Date: 2026-06-17 · **Hardware:** 8 GB consumer GPU (~7.3 GB free), 10 CPU threads **Method:** real ELI sessions (quick mode, full persona+memory pipeline), not lab benchmarks.

Read this first — honest caveats

- These are **single informal sessions**, one per model, on one machine — *directional*, not controlled.
- Every conversational turn carries ELI's **6-8 k-token persona + memory brief**; on this hardware **prompt-eval (prefill) dominates the turn time**, not generation. That brief is ELI's depth and is *not* a thing to trim — so treat turn-times as “what living with this model felt like,” not tok/s.
- Background daemon work (news synthesis, self-improvement) **interleaved** with foreground turns in some logs, inflating them. Several issues seen here are **already fixed in code** (flagged ■ below); the logs predate those fixes.
- Counts/timings below are quoted from the actual run logs.

1. The contenders (with live load facts)

Model	File size	Arch	Active/token	Thinks?	Loaded as (smart-fit)	n_ctx_train
Qwen3.6-35B-A3B (baseline)	21 GB	MoE transformer	~3 B of 35 B	Yes	ctx 20480, gpu 8 , batch 64	262144
Falcon-H1-34B-Instruct	18.94 GB	Dense hybrid (attn+SSM/Mamba)	all 34 B	No	ctx 20480, gpu 8 , batch 64	262144
Qwen3-8B	4.68 GB	Dense transformer	all 8 B	Yes	ctx 26624, gpu 27/36 , batch 128	40960
Phi-4 (14.7 B)	8.43 GB	Dense transformer	all 14.7 B	No	ctx 28672, gpu 16/40 , batch 128	16384

2. Headline comparison

Dimension	A3B (35B MoE)	Falcon-H1-34B	Qwen3-8B	Phi-4 (14B)
GPU residency	8 layers (bad)	8 layers (bad)	27/36 (good)	16/40 (partial)
Routing JSON call	~30–80 s	81 s	4.5 s	19–38 s (fenced) ▮
Conversational turn	147–537 s (+empty-loops)	736–839 s	48–90 s	64–177 s
Bare commands (pause/vol/next)	instant	instant	instant	instant
Thinking empty-loops	Catastrophic (30+ min)	None	None (this run) ▮	None
Fabricated actions	some confab	minimal (6-turn run)	none seen	“[Pause executed]” + fake file analysis sycophantic mirror overshoot (28672 > 16384)
Persona / voice	strong but slow	dry, witty (small sample)	most original	sycophantic mirror overshoot
ctx vs trained	undershoot (32k vs 262k)	undershoot	ok	(28672 > 16384)
Verdict on this card	retired	not viable	best fit	capable backup

3. Per-model breakdown

3.1 Qwen3.6-35B-A3B (the baseline that started all this)

A 35 B Mixture-of-Experts (~3 B active). On paper the active-param count should be fast; in practice the **whole 35 B must sit in memory**, so on 8 GB it ran heavily CPU-offloaded at ~2 tok/s, “lumpy.”

- **Thinking model → the empty-<think> disaster:** introspection turns burned **30+ minutes** across stacked ~530 s generations that returned *empty strings* (budget eaten inside <think>). This is the failure that motivated the governor work.
- Confabulation under stress (the “job #3 is a ghost”, LIST_NOTES discarded, fabricated status).
- **Status:** superseded. The MoE memory pattern + thinking-loops make it the wrong tool here.

3.2 Falcon-H1-34B-Instruct — the slowest, by design

Dense hybrid (Transformer attention + Mamba/SSM), **all 34 B active every token**. On an 8 GB card with only 8 layers offloaded, that’s the worst case.

- “How’s the head?” took **~8 minutes** (81 s just to *route*, 413 s to generate).
- The “model agnosticism” turn hit **839 s (~14 min)** — and the log shows *why*: a **340 s news synthesis** + a **117 s self-improvement** call sitting on the shared model lock mid-turn ▮, plus the self-improvement loop firing **10+ times (~100 s each)** in the background, flooding the run.
- **Plus:** non-thinking (no empty-loops), and the SSM layers give better long-context memory scaling — genuine architectural pluses that don’t matter when each token costs 34 B of compute on this GPU.
- **P vs NP** answer was honest and witty (good quality), but the latency is disqualifying here.
- **Status:** not viable on 8 GB. Revisit on a bigger card, where the hybrid is interesting.

3.3 Qwen3-8B — the winner on this hardware

4.68 GB, 27 of 36 layers on GPU → the most resident, and it shows.

- **Fastest turns: ~48–90 s**, routing in **4.5 s**. Bare commands instant (PLAY 1.2 s, VOLUME/NEXT <0.1 s).
- **Best persona:** the birthday exchange had genuine character and originality (it pushed back, it had a voice) — this is the model that most sounds like *ELI*.
- **Thinking model**, but it ran **clean this session** — no empty-loops observed. With the empty-<think> retry now in code ▮, the thinking risk is largely neutralised.
- **One load-policy**

quirk: smart-fit first tried *all* layers at ctx 20480 → **failed** → fell back to 27 layers + a *bigger* ctx 26624. That's backwards for speed: it kept a large context and offloaded 9 layers. Dropping ctx to ~10-12 k would likely fit **all** layers and make it noticeably faster (a load-policy fix, not a model flaw). - **Status: recommended default** for daily ELI on this card — speed *and* voice.

3.4 Phi-4 (14.7 B) — capable, but the integrity problems are real

8.43 GB, 16 of 40 layers on GPU. - **Turns ~64-177 s** — *slower than the smaller Qwen3-8B*, because less of it is resident. - **Non-thinking** (no empty-loops) — its main genuine advantage. - **Three integrity issues, all visible in the log:** 1. **Fabricated actions** — “Pausing Spotify now. **[Pause command executed successfully.]**” with no pause executed; and a confidently **fabricated analysis of two files it never opened** (“holy war diabolic”). You caught it (“do not lie”). ▶ *No-fake-action strip now in code.* 2. **Fenced JSON** — Phi wraps routing JSON in ``json ... ``, and a long-path JSON exceeded the 90-token routing cap → unparseable → fell to CHAT → fabrication. ▶ *Fence-strip + bigger cap now in code.* 3. **Sycophantic mirroring** — it restates your input with light rephrasing rather than bringing its own voice. You flagged this directly; it's **model-intrinsic** (not a regex fix). - **ctx overshoot:** loaded at ctx 28672 while its trained context is **16384** → “possible training context overflow.” ELI over-asked here (the mirror of the A3B's undershoot) — both argue for reading `n_ctx_train` from metadata (governor item). - **Status:** solid **non-thinking backup**; but on this card it's slower than Qwen3-8B and needs the integrity guards to behave.

4. The cross-cutting findings (true across the field)

1. **Bare commands are flawless on every model** — the deterministic fast-path (PHASE45) executes pause/play/next/volume in <0.1-1.7 s regardless of model. ELI's command spine is genuinely model-agnostic. [yes]
 2. **Latency is dominated by the 6-8 k-token brief**, not the model's raw speed. On weak hardware, “smaller model fully resident” beats “bigger model offloaded” every time (Qwen3-8B < Phi-4 < Falcon).
 3. **The context heuristic is wrong in both directions** — undershoot for A3B (32 k vs 262 k trained), overshoot for Phi-4 (28 k vs 16 k trained). Prefer the model's own `n_ctx_train` metadata.
 4. **The proactive daemon was the hidden latency multiplier** — news synthesis + self-improvement stealing the shared lock mid-turn (worst on Falcon: 340 s news block). ▶ *Daemon-deferral + news background-flagging now in code.*
 5. **Thinking vs non-thinking is the key architectural axis for ELI:** thinking models (A3B, Qwen3) risk the empty-<think> failure (now guarded ▶) but tend to richer voice; non-thinking models (Falcon, Phi) avoid it but Phi shows more fabrication/sycophancy. No free lunch.
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5. Recommendation (per ELI's values: integrity + depth over raw speed)

Daily driver: Qwen3-8B. Fastest on this card, most original voice, clean run — and the empty-<think> guard removes its only structural risk. To squeeze it further, fix the load policy so it goes *fully* GPU-resident (smaller ctx, all layers).

Backup: Phi-4, for any session where you want a non-thinking model — but only with the no-fake-action and fenced-JSON guards active (now in code), because its fabrication/yes-man tendencies are real.

Shelved on this hardware: Falcon-H1-34B and the 35B-A3B — both too slow (dense-34B and offloaded-MoE respectively). Re-test both on a bigger GPU, where Falcon's SSM hybrid becomes genuinely interesting.

Net: you called Phi-4 the “goldilocks,” but the side-by-side actually nudges toward **Qwen3-8B** — it's *faster and more ELI* than Phi-4 here, and the thinking-loop reason to prefer Phi is now handled in code.

6. Appendix — untested candidates already in models/

(No run data; predictions from architecture + this bake-off. Listed for completeness.)

Model	Size	Prediction on 8 GB
Qwen3-32B-Q4_K_M	19 GB	Dense 32 B → Falcon-class slow. Shelf until bigger GPU.
Qwen3.6-27B-UD-Q4_K_XL	17 GB	Dense 27 B → slow, offloaded. Shelf.
Qwen2.5-7B / mistral-7b-v0.2 / openhermes-7b	~4-5 GB	Would be fast (resident) but older than Qwen3-8B; lower quality, non-thinking. Only if you want a no-think 7 B.
ministral-3b / phi-2 / tinyllama	<3 GB	Very fast, but too small for ELI's reasoning/persona depth.
DeepSeek-R1-Distill-Qwen-1.5B	~1 GB	Tiny <i>thinking</i> model — fast but empty-loop-prone and too small.
Qwen2.5-VL-7B	4.4 GB	Vision model (used for image/screen), not a text driver.

7. Fix status referenced above (🚩 = landed in working tree, uncommitted)

- Empty-<think> retry (force no-think on empty) — inference_broker.py 🚩
- Background-work deferral when foreground is live — inference_broker.py 🚩
- News synthesis flagged background (capped/preemptible/deferred) — proactive_daemon.py 🚩
- Fenced/truncated routing JSON tolerance — llm_intent.py 🚩
- No-fake-action tool-confirmation strip — output_governor.py 🚩
- (Earlier, committed a641471) job-status routing, CREATE_FILE exec, LIST_DIR capture, multi-command banter, compact-synthesis no-think.

Still open (model-intrinsic or deferred): Phi-4 sycophantic tone (model choice), command-buried-in-chat extraction, n_ctx_train-aware context sizing, load-policy “full-GPU for small models”.